

Cadence Design Systems, Inc.

CDNS | October 4, 2026

HOLD

Fair value (12-mo horizon reference)	\$308.00
Current price (October 2, 2026 close)	\$351.35
Implied upside / downside	-12%
Verdict	HOLD

Investment Thesis

Cadence is the purest software toll on the AI hardware buildout, and its business quality is beyond dispute: 15 consecutive years of return on invested capital above 15% (2011-2025), gross margins that have expanded steadily into the high-80s on the software mix, free-cash-flow conversion above 80%, and a record \$8.1 billion backlog with \$4.2 billion converting to revenue within twelve months. Second-quarter 2026 revenue grew 24.2% to \$1.584 billion, non-GAAP operating margin hit 45.5%, and management raised full-year guidance to \$6.26-\$6.34 billion of revenue and \$8.05-\$8.15 of adjusted EPS. Every chip that matters - AI accelerators, hyperscaler silicon, leading-edge mobile - is designed on Cadence or Synopsys tools. This is a structural winner, and we treat it as one.

But structural winners get priced as though structure means certainty, and the current price of \$351.35 embeds a growth trajectory that the semiconductor cycle has never once delivered smoothly. Cadence's revenue is ultimately a function of semiconductor R&D; spending, which is cyclical with a capital-S: customers' R&D; budgets expand in booms and get scrutinized in downturns, EDA spending lags chip downturns by a few quarters, and the current AI-driven R&D; supercycle is concentrating growth in a handful of hyperscaler and AI-accelerator customers whose spending plans can pivot. The stock has already corrected about 32% from its 52-week high of \$416.69 - the market is telling you something about how it prices the cycle - and even after that correction, the multiple assumes the AI R&D; boom extends deep into the decade.

Our judgment, on a probability-weighted scenario DCF with the 15-year compounder-quality treatment the company's history has earned (8-8.5% base discount rate, 3.0% terminal growth cap on demonstrated sustained margins), is fair value of \$308 - about 12% below the current price. The base case is genuinely bullish on the business: EDA grows as a share of semis R&D;., agentic AI design tools (ChipStack, ViraStack, AuraStack) expand consumption of Cadence's engines rather than cannibalizing seats, the IP and System Design & Analysis businesses keep compounding at 30%+, and margins stay in the mid-40s. The bear case - which sits below the price, as required - models the R&D; cycle turning: hyperscaler silicon programs pause, China export controls bite harder, and the multiple compresses as growth decelerates. We rate the stock

HOLD: own the compounder, but do not chase it 12% above fair value into a cyclical peak in customer R&D; spending.

Business Overview

Cadence Design Systems is one of two dominant electronic design automation (EDA) vendors - the other being Synopsys - providing the software that the semiconductor industry uses to design, verify, and simulate chips and electronic systems. The product portfolio spans Core EDA (digital and custom/analog design, including the Virtuoso, Innovus, and Genus platforms), Functional Verification (the Palladium hardware emulation family), semiconductor IP (memory, interface, and connectivity blocks), and System Design & Analysis (multiphysics simulation for 3D-IC, thermal, and electromagnetic effects). Roughly three-quarters of revenue is recurring, and the backlog-driven model gives unusual forward visibility.

The financial profile is elite-software economics. Second-quarter 2026 revenue of \$1.584 billion grew 24.2% year over year, with Core EDA up 18%, IP up more than 40%, and System Design & Analysis up 37% (helped by the BETA CAE and Hexagon Design & Engineering acquisitions). Non-GAAP operating margin was 45.5%, free cash flow was \$582 million in the quarter, and the company ended June with \$1.44 billion in cash. Customers include Nvidia, Apple, and effectively every leading-edge chipmaker and systems company. The TSMC relationship deepened through 2026 with tool certifications across the A14 (1.4nm) and N2P nodes and expanded 3DFabric support - table stakes in the foundry-duopoly world, but Cadence holds its seat at the table.

The strategic frontier is AI for design. Cadence's agentic tools - ChipStack, ViraStack, and the AuraStack 'super agent' launched in 2026 - let engineers describe design goals in plain language while the agents run hundreds of parallel iterations on Cadence's physically accurate engines. Management's disclosed insight is important: autonomous agents increase consumption of the underlying EDA engines rather than cannibalizing seat licenses, because every agent iteration burns compute on licensed solvers. If that dynamic holds, agentic AI is a demand accelerator for the core franchise, not a disruption of it. The offsetting overhangs: the Hexagon D&E; integration is diluting near-term margins (roughly \$0.28 of 2026 EPS impact), China export-control policy remains a swing factor for a meaningful revenue geography, and Synopsys - now larger after closing the Ansys acquisition in July 2025 - is a well-armed competitor across the full stack.

Forward Outlook

Our forward view is that Cadence grows revenue in the mid-teens through the decade in the base case, driven by three durable forces: AI-accelerator and hyperscaler silicon programs expanding the leading-edge design-start pool, EDA's share of semiconductor R&D; spend continuing its decades-long creep upward as complexity rises, and the IP and multiphysics businesses compounding faster than core EDA. The \$8.1 billion backlog - more than a full year of revenue, with \$4.2 billion converting in twelve months - underwrites the near term almost mechanically. This is why the business earns the compounder-quality assumptions: the revenue visibility is contractual, the switching costs are measured in years of methodology investment, and no startup is displacing a duopoly entrenched in every leading fab's reference flow.

The cycle is the caveat, and it is a large one. Semiconductor R&D; spending is growing at an extraordinary rate because a small number of very large customers are in an arms race; arms races end, pause, or consolidate. When they do, EDA feels it with a lag but feels it fully - design starts get pushed, IP licensing slows, and hardware emulation purchases defer. Our base case assumes the current R&D; supercycle moderates rather than collapses, with growth stepping down from the 20%+ of 2026 toward the low teens by the end of the decade. We do not assume a semiconductor downturn in the base case, which is itself a generous assumption

given the industry's history of one every few years.

On agentic AI we take management's demand-accelerator claim seriously but haircut the monetization timeline. The mechanism - agents burning licensed solver compute - is economically sound, but pricing models for agent-driven consumption are still being invented, and customers will negotiate. The Hexagon integration should be accretive by 2027-2028 as cost synergies land and the multiphysics portfolio cross-sells into the core EDA installed base. Net: a business we would love to own at the right price, growing into a valuation that currently assumes the R&D cycle has been repealed.

Valuation

Cadence qualifies for our compounder-quality treatment: 15 consecutive years of ROIC above 15% (2011-2025), stable-to-expanding gross margins, and free-cash-flow conversion above 80%, all verified from reported history. We therefore model a 15-year explicit horizon at an 8-8.5% base discount rate, with terminal growth capped at 3.0% applied to a demonstrated sustained margin - not a cyclical peak in customer R&D spending. The bear case adds 250bp, the bull case subtracts 150bp with an 8% floor, weighted 25% / 50% / 25%. Terminal value exceeding 70% of enterprise value is haircut and disclosed.

The bear case, sitting below the current price, assumes the AI R&D supercycle rolls over: hyperscaler silicon programs consolidate, a semiconductor downturn arrives mid-decade, China revenue is impaired by export controls, and revenue growth falls to high-single digits with margins compressing as the Hexagon integration disappoints. The base case assumes mid-teens revenue growth moderating to low teens, non-GAAP operating margins sustained in the mid-40s, agentic tools expanding engine consumption as management describes, and the backlog converting on schedule. The bull case assumes the AI silicon buildout extends through the decade, Cadence's agentic platform becomes the industry-standard design cockpit with premium pricing, and EDA's share of R&D spend steps up structurally.

The probability-weighted fair value is \$308.00, our target, against the October 2, 2026 close of \$351.35 - about 12% downside. The valuation respects the franchise fully - the 15-year horizon and 8-8.5% discount rate are the most favorable assumptions we grant any business - and still finds the price ahead of the fundamentals. The stock's own 32% drawdown from its high is the market's way of saying the cycle is being repriced; our work says the repricing is not finished.

Key Risks

- Semiconductor R&D; cyclical: EDA revenue lags chip downturns but follows them; a pause in hyperscaler silicon spending would hit design starts, IP licensing, and hardware sales with a lag.
- Customer concentration: AI-accelerator and hyperscaler programs are a growing share of growth; consolidation or insourcing shifts by a few large customers would be material.
- China export controls: China is a meaningful revenue geography and policy is a swing factor outside the company's control.
- Hexagon integration: the Design & Engineering acquisition is diluting 2026 margins; synergy capture and cross-sell execution must deliver to justify the price paid.
- Synopsys competition: the Ansys combination makes the arch-rival larger and broader across simulation; share battles in verification and IP could pressure pricing.

- Agentic-AI monetization: if agent-driven consumption commoditizes solver pricing or shifts value to the agent layer, the demand-accelerator thesis weakens.
- Valuation: even after the drawdown, the multiple assumes a smooth continuation of 20%+ growth that the semiconductor cycle has never delivered.

What Would Change Our Mind

- A deeper cyclical washout - the stock offered 20%+ below our fair value on a semiconductor downturn that leaves the duopoly intact - would make this a BUY; that is historically when Cadence should be bought.
- Proof that agentic tools are expanding per-customer spend at premium pricing (not just shifting seats to consumption) would raise our through-cycle growth rate.
- Sustained 20%+ revenue growth with 45%+ non-GAAP margins through an actual semiconductor downturn would prove the business has de-cyclicalized and justify a higher multiple.
- Conversely, ROIC falling below 15% for consecutive years, structural gross-margin compression, or FCF conversion dropping below 80% over a multi-year window would revoke the compounder-quality assumptions and lower fair value.
- Loss of the TSMC leading-node certification cadence or a major foundry reference-flow defection would impair the moat directly.

Valuation Approach

Our valuation is a probability-weighted scenario DCF. Cadence meets our compounder-quality criteria, verified from history rather than projected: a 15-year streak (2011-2025) of return on invested capital above 15%, stable-to-expanding gross margins across the cycle, and free-cash-flow conversion above 80%. This qualification is not permanent: if ROIC were to fall below 15% for consecutive fiscal years, if gross margins compressed on a structural rather than cyclical basis, or if FCF conversion dropped below 80% over a multi-year window, the qualification would be revoked and the business re-valued at the standard 10% discount rate over a 10-year horizon. Under the qualification we model bear, base, and bull cases over an explicit 15-year forecast horizon, weighted 25% / 50% / 25%, with a base discount rate of 8-8.5% reflecting the genuinely lower risk of predictable cash flows. The bear case adds 250bp and the bull case subtracts 150bp (floor 8%). Terminal value assumes no more than 3.0% perpetual growth applied to a demonstrated sustained margin level - not a peak - and any terminal value exceeding 70% of enterprise value is haircut and disclosed. Bear cases are required to be genuinely adverse and to sit below the current price. Management guidance is never accepted at face value; it is independently tested and haircut where evidence warrants.

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